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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS – AUGUST / SEPTEMBER 2023

Program	:	**B.E. - Information Science and Engineering**	**Semester**	:	**IV**
Course Name	:	**Database Management Systems**	**Max. Marks**	:	**100**
Course Code	:	**IS45**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1. a) State and explain the three schema architecture of DBMS. Why do you need mappings between the different levels of architecture? CO1 (08)

   b) Compare DBMS with early File system and list any six advantages of using DBMS approach. CO1 (06)

   c) What are the responsibilities of the DBA and a database designer? CO1 (06)

2. a) Define the following. CO1 (05)
   i) DDL ii) DML iii) Data Model iv) Weak entity v) SDL.

   b) Discuss the main categories of data models. CO1 (07)

   c) Develop an ER diagram for keeping track of information about a BANK database. Each bank can have multiple branches and each branch can have multiple accounts and loans. Indicate all key and cardinality constraints. CO1 (08)

UNIT - II

3. a) Given the schema: CO2 (07)

   Student (<u>USN</u>, Name, Branch, Percentage)
   Faculty (<u>FID</u>, Fname, Dept, Designation, Salary)
   Course (<u>CID</u>, Cname, FID)
   Enroll (<u>CID, USN</u>, Grade)

   Give the **relation algebra** expression for the following:

   i. Retrieve the name and percentage of all students for the course IS435
   ii. List the departments having average salary of the faculties above Rs 45,000

   b) Suppose that each of the following update operations is applied directly to the database state shown below. Discuss all integrity constraints violated by each operation, if any, and the different ways of enforcing these constraints. SSN and Dnum are primary keys. CO2 (06)

Emp

FName	MINT	LNAME	<u>SSN</u>
John	B	Smith	1234

Dept

DName	<u>Dnum</u>	MgrSSN
Research	5	1234

i) Insert <'Robert', 'F', 'Scott', null> into Emp
ii) Insert <'Admin', 5, 1234> into Dept
iii) Delete the EMP tuple with SSN=1234
iv) Update <Research, 5, 1222> into Dept where MgrSSN = 1234.

c) Consider the following two tables T1 and T2. Show the results of the following operations. (Assume T1 and T2 are union compatible) CO2 (07)

Table T1

P	Q	R
10	a	5
15	b	8
25	a	6

Table T2

A	B	C
10	b	6
25	c	3
10	b	5

i) $T_1 \bowtie_{T_1 \cdot P = T_2 \cdot A} T_2$

ii) $T_1 ⟕_{T_1 \cdot P = T_2 \cdot A} T_2$

iii) $T_1 \bowtie_{T_1 \cdot P = T_2 \cdot A \text{ AND } T_1 \cdot R = T_2 \cdot C} T_2$

iv) $T_1 \cup T_2$

v) $T_1 ⟖_{T_1 \cdot Q = T_2 \cdot B} T_2$

4\. a) Consider the following relations for a database that keeps track of business trips of salespersons in a sales office: CO2 (05)

SALESPERSON (SSN, Name, Start_Year, Dept_No)
TRIP (SSN, From_City, To_City, Departure_Date, Return_Date, Trip_ID)
EXPENSE (Trip_ID, Account#, Amount)

Specify the primary and foreign keys for this schema, stating any assumptions you make.

b) Given the schema EMP (Fname, Lname, SSN, Bdate, Address, Sex, Salary, SuperSSN, Dno) CO2 (09)
DEP T(Dname, Dnumber, MgrSSN, MGrstartdate)
DEPT-LOC (Dnumber,Dloc)
PROJECT(Pname, Pnumber, Ploc,Dnum)
WORKS-ON (ESSN,PNo,Hours)
Give the **relation algebra** expression for the following:

i. List female employees from Dno=20 earning more than 50000
ii. Retrieve the first name, last name and salary of all employees who work in department number 50.
iii. Retrieve the name of the manager of each department.

c) List the steps involved in ER-to-Relational mapping. CO2 (06)

UNIT - III

5\. a) Consider the following table with the details of products sold in a shop for two days. CO3 (09)

PRODUCTS

Date	ItemName	Unit_Sold
9/3/23	Pencils	100
9/3/23	Pen	500
10/3/23	Pencils	50
10/3/23	Pen	900
10/3/23	Eraser	75

Construct SQL queries for the following scenarios:

i) Display the total number of items sold for each item name.
ii) Display the item names and total number of items sold for only those items where total of sold items is greater than 1000.
iii) Display the table sorted in the descending order on the number of quantities sold.

b) Explain GROUP BY function in SQL with an example. CO3 (05)

c) Specify three different ways of inserting value into the table. Create a table in SQL which fulfil the following criteria's: CO3 (06)

i) Specifying attribute constraints

ii) Specifying key and referential constraints.

6. a) **Employee** (Name, SSN, address, Sex, salary, DNo) CO3 (10)
Department(Dname, Dnumber, MGRSSN, MGRSTARDATE)
Dept_locations (Dnumber, Dlocations)
Project (Pname, pnumber, plocation, Dnum)
Work_on (ESSN, PNO, Hours)
Dependent (ESSN, Dependent_name, sex, B_date, Relationship)
Write the queries in SQL to

i. Retrieve the name and address of all the employees who work for the "research" department.
ii. Find the names of the employee who works on all projects controlled by dnumber 5.
iii. List all the projects on which the employee "smith" is working.
iv. Retrieve the names of the employee who have no dependents.

b) How does SQL allow implementation of the entity integrity and referential integrity constraints? Give one example for each. CO3 (10)

UNIT- IV

7. a) Explain with an example for the Informal guidelines for a relation schemas. CO4 (07)

b) What is the need for Normalization? Explain second normal form. Consider the relation EMP-PROJ= {SSN, Pnumber, Hours, Ename, Pname, Plocation}. Assume {SSN, Pnumber} as primary key. The dependencies are: CO4 (07)
SSN Pnumber → {Hours}
SSN→ {Ename}
Pnumber → {Pname, Plocation}
Normalize the above relation into 2NF.

c) List the Inference rules for functional dependencies. Write the algorithm to determine the closure of X (set of attributes) under F (set of functional dependencies). CO4 (06)

8. a) Discuss insertion, deletion and modification anomalies. Why are they considered bad? Illustrate with examples. CO4 (07)

b) If the relation contains multi valued attribute what are different techniques to achieve 1NF. Illustrate with an example and also find which technique is the best. Justify your answer. CO4 (06)

c) Explain BCNF. State if the following relation is in BCNF. If not, decompose the same into BCNF. State the candidate key. CO4 (07)

STUDENT	COURSE	INSTRUCTOR

FD1: {Student, Course} -> Instructor
FD2 -> Instructor -> Course

UNIT - V

9.	a)	Explain properties of transaction with state transaction diagram.	CO5	(06)
	b)	Discuss Recovery techniques based on Deferred Update.	CO5	(08)
	c)	Why NoSQL? List the characteristic of NoSQL database.	CO5	(06)
10.	a)	What do you understand by a transaction? In what situation a transaction is said to be committed or aborted?	CO5	(08)
	b)	What is the Need of concurrency control and recovery?	CO5	(06)
	c)	Describe the features of MongoDB.List the Datatypes with an example for each.	CO5	(06)

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